

PRACTICE EXAM

Difficulty: MEDIUM

Questions: 10

Electrical Engineering and Python GUI Exam

Multiple Choice Questions (1 point each)

Instructions: Choose the best answer for each question.

Question 1: According to the Introduction section, what is one of the concepts this lab is meant to reintroduce?

- A) Fourier Transform
- B) Superposition Principle
- C) Thevenin's Theorem
- D) Norton's Theorem

Question 2: In the circuit under test, what component is in series with the resistors on the left side?

- A) Inductor
- B) Diode
- C) Capacitor
- D) Voltage Source

Question 3: Based on the PyQt5 vs PyQt6 comparison, what is the main reason to use PyQt6?

- A) It has better performance
- B) It is more widely used
- C) It is more modern
- D) It is compatible with older systems

Question 4: Which PyQt6 module contains the QMainWindow class?

- A) QtCore
- B) QtGui
- C) QtWidgets
- D) GStreamer

Short Answer Questions (3 points each)

Instructions: Answer each question in 2-3 sentences.

Question 5: Briefly describe the objective of the electrical engineering lab described in the text.

Question 6: Explain what the superposition principle states, based on information from the provided text.

Question 7: Describe what a QSplitter is in the context of PyQt6 and give an example of how it can be used in a GUI.

Problem-Solving Questions (5 points each)

Instructions: Provide detailed answers, showing your reasoning and application of concepts.

Question 8: Considering the experimental results in Tables E1, E2, and E3, discuss the effect of the capacitor (open, shorted, and present) on the output RMS voltage (v_O (RMS)). What trends do you observe?

Question 9: Explain how you might recreate the aspect ratio of a video feed within PyQt, given the current `Gtk.AspectFrame` implementation as described in the provided text.

Question 10: Suppose you are designing a PyQt application to display multiple HD video streams using GStreamer. Which of the two GStreamer video rendering options described ("VideoOverlay into a QWidget" or "appsink -> QImage -> QLabel/QOpenGLWidget") would you choose and why? Justify your choice based on performance and flexibility considerations.